Aramis Tanelus

Contact Information

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LinkedIn https://www.linkedin.com/in/arajt
GitHub https://www.github.com/Aramist

Education

New York University New York, NY
BS, Neural Science Aug 2019 – Dec 2023
Ph.D., Data Science Sep 2025 – Present

Experience

NYU Music and Audio Research Lab New York, NY
Ph.D. Student – Dr. Brian McFee Sep 2025 – Present

- Currently investigating the geometry of embeddings generated by large pretrained audio models with the goal of understanding how sensitivities to different qualities of the input audio are introduced to the models and whether we can control this.

Flatiron Institute New York, NY
Research Analyst (current), Guest Researcher (former), Intern (Former) Jun 2022 – Aug 2025

- Explored the use of generative models to study the relationships between vocal activity and social interactions in Mongolian Gerbils (*M. unguiculatus*).
- Developed deep neural networks to localize rodent vocalizations captured in reverberant environments and provide estimates of predictive uncertainty.
- Worked across labs and recording environments to build and publish a tool which could be easily adopted by other social neuroscience researchers.
- Volunteered as a teaching assistant for workshops held by the Center's Neuro Research Software Engineer group in Vienna, AT and New York, NY. Assisted researchers from around the world in assignments designed to foster proficiency in open-source tools for loading and manipulating data types common to computational neuroscience research.
- Active member of the Center's coding club, which is an informal venue for researchers in the Center for Computational Neuroscience to teach others about useful packages and best practices.

New York University New York, NY
Research Assistant – Dr. Dan Sanes and Dr. David Schneider Lab Jun 2021 – May 2023

- Developed a system to conduct long-term, continuous, synchronized recordings through multiple high-bandwidth microphones and cameras using National Instruments hardware.
- Analyzed ultrasonic Mongolian Gerbil (*M. unguiculatus*) vocalizations and corresponding electrophysiological responses in the primary auditory cortex.

New York University Tandon School of Engineering Brooklyn, NY
Calculus Tutor – TRIO Scholars Program Sep 2020 – May 2021

- Taught Calculus I and II in a one-on-one setting to students in the TRIO Scholars program, a chapter of the U.S. DoE's Student Support Services program aimed at supporting modest-income, first-generation, and disabled students in STEM fields.
- Created videos intuitively explaining calculus concepts and demonstrating practice problems to supplement weekly lectures.

LexisNexis Risk Solutions Boca Raton, FL
Intern May 2018 – Aug 2018

- Worked alongside a mentor to design and implement a library to connect robotics engineers with HPCC Systems' parallel computing services.
- Gained exposure to methods of storing and querying large volumes of data.
- Presented a poster on this project at the 2018 HPCC Community Day Summit.

Awards

McKnight Foundation
MEFN Matthew Pecot Fellowship Jun 2022 – May 2023

- Awarded $16,500 to support my research as an undergraduate student and invited to attend the 2023 McKnight Conference on Neuroscience.

Publications

- Peterson RE*, **Tanelus A***, Ick C, Mimica B, Muttath Joseph NF, Ivan V, Choudhri A, Falkner A, Murthy M, Schneider DM and Sanes DH, 2024. Vocal Call Locator Benchmark (VCL) for localizing rodent vocalizations from multi-channel audio. Advances in Neural Information Processing Systems, 37, pp.106370-106382. https://vclbenchmark.flatironinstitute.org/
- Peterson RE, Choudhri A, Mitelut C, **Tanelus A**, Capo-Battaglia A, Williams AH, Schneider DM, Sanes DH. "Unsupervised discovery of family specific vocal usage in the Mongolian gerbil." *eLife* (2023)

Talks

- **Tanelus A**, Mimica B, Francis N, Falkner AL, Murthy M, Schneider DM, Sanes DH, Williams AH. "Weakly Supervised Vocal Call Attribution" Presented at:
  - *Gordon Research Seminar: Neural Mechanisms of Acoustic Communication*, May 30 – 31, 2026

Posters (presenting authors underlined)

- **Tanelus A**, Peterson RE, Choudhri A, Ick C, Ivan V, Francis N, Schneider DM, Sanes DH, Williams AH. "Benchmarks and Calibrated Neural Networks for Localizing Vocalizations in Social Interactions." Presented at:
  - *Gordon Research Conference: Neural Mechanisms of Acoustic Communication*, May 19 – 24, 2024

- Peterson RE, **Tanelus A**, Choudhri A, Ivan V, Prasad A, Schneider DM, Sanes DH, Williams AH. "Benchmarks and deep learning models for localizing rodent vocalizations in social interactions." Presented at:
    - *Neural Information Processing Systems (NeurIPS) Machine Learning for Audio Workshop*, Dec 16, 2023
- *Peterson RE, **Tanelus A**, Choudhri A, Ivan V, Prasad A, Kirchgessner M, Magland J, Soules J, Froemke RC, Schneider DM, Sanes DH, Williams AH. "Deep sound source localization for the study of social vocal interactions in rodents." Presented at:*
    - *Advances and Perspectives in Auditory Neuroscience*, Nov 10, 2023
    - *Society for Neuroscience*, Nov 11 – 15, 2023
- Peterson RE, **Tanelus A**, Choudhri A, Ivan V, Schneider DM, Sanes DH, Williams AH. "Towards the neuroethology of vocal communication in the Mongolian gerbil." Presented at:
    - *Computational and Systems Neuroscience (COSYNE)*, Feb 29 – Mar 5, 2024
- Peterson RE, Mitelut C, **Tanelus A**, Choudhri A, Yamaguchi A, Capo-Battaglia A, Williams AH, Schneider DM, Sanes DH. "The spectrotemporal organization of familial gerbil vocalizations." Presented at:
    - *International Conference on Auditory Cortex*, Sep 2022
    - *Gordon Research Conference: Neural Mechanisms of Acoustic Communication*, Aug 2022
- Peterson RE, Mitelut C, **Tanelus A**, Capo-Battaglia A, Schneider DM, Sanes DH. "Organization of Mongolian Gerbil Social Vocalizations." Presented at:
    - *Association for Research in Otolaryngology,* Feb 5 – 9, 2022